Writing a program

The shape of a Test1 program, and the one rule that keeps it readable a year later.

A program is a list of lines the controller runs top to bottom, ten times a second. That is the whole model: no tasks, no threads, and no reason to fear it.

The shape of a program

```text
10  IF IN1 = ON THEN OUT1 = ON
20  IF IN1 = OFF THEN OUT1 = OFF
30  IF AI1 > 22.5 THEN OUT2 = ON
```

Every program worth keeping has three parts: what it reads, what it decides, and what it writes. Put them in that order and a stranger can read it. Mix them and only the person who wrote it can.

The rule

One output is set in one place. Two lines that both write `OUT1` will, sooner or later, disagree — and the machine will do something nobody intended at a moment nobody predicted. If two conditions can drive one output, write one line that decides between them.

Download and watch

Downloading replaces the running program. The controller keeps running its old program until the download completes, then switches — so a download is not a gap in control, and a failed one leaves the old program in place.

Watch the first ten seconds after a download. A program that was going to misbehave almost always shows it in the first scan of the inputs.